

Data Mining and Visualization (83676)

Final Project

Due: 24/6/26

General

In this project you will apply in Python the concepts you have learned in the course. The main tasks are to analyze the given dataset, apply preprocess techniques and get a prediction using classification methods.

Submission is in threes only!

The submission should include three files:

- Report: a PDF file with the project description and all the explanations of the process and the results in English or Hebrew.
- A Python notebook file (.ipynb) that has the full code and **all the results displayed** in the same order as in the report. Missing results and messy or unreadable code may lead to a lower score.
- A CSV file with predictions for the attached test dataset. It should contain only the predicted values in the same order as the test dataset.

The grade evaluation takes into consideration:

- Techniques used: Did you select appropriate and diverse techniques and justify why?
- The process you followed: Is it correct (given the techniques you used), did you describe it well?
- Interpretation of results: Did you correctly understand and interpret the results you obtained?
- Quality of writeup: Did you present your work well, in an understandable and usable manner?
- The classification model quality: The test set predictions will be ranked based on **accuracy** relative to the class, this part will constitute 10 points of the grade (those ranked at the bottom will receive 1 points).
- Missing parts or explanations may lead to a lower score.

Problem Description

Understanding user behavior in online platforms is essential for improving conversion rates and optimizing marketing strategies. While many users visit e-commerce websites, only a portion demonstrate meaningful engagement or intent to perform valuable actions.

In this project, you will analyze a dataset of user sessions from an online shopping platform. The dataset includes behavioral and contextual information such as browsing activity, session characteristics, and user attributes. The objective is to develop a predictive model that identifies **high-intent sessions**, i.e, sessions that indicate a strong likelihood of valuable user interaction or conversion.

Accurate prediction of high-intent sessions enables companies to target users more effectively, improve decision making, and enhance overall business performance.

Data Attributes

The dataset contains information about user sessions, including browsing behavior, interaction patterns, and contextual features. Each row in the dataset represents a single user session, capturing the user behavior and interaction with the website during one visit.

- **num_admin_pages:** Number of administrative pages visited during the session (e.g., account or settings pages).
- **admin_duration:** Total time spent on administrative pages during the session.
- **num_info_pages:** Number of informational pages visited (e.g., help pages, policies, or general information).
- **info_duration:** Total time spent on informational pages.
- **num_product_pages:** Number of product-related pages visited during the session.
- **product_duration:** Total time spent browsing product-related pages.
- **bounce_rate:** The proportion of interactions in the session where the user leaves after viewing only a single page. Higher values indicate low engagement.
- **exit_rate:** The tendency of a session to end after visiting certain pages, reflecting how often users leave the website during the session.
- **page_value:** An estimate of the commercial value associated with the pages viewed during the session, based on historical users' behavior.
- **special_day_score:** Indicates proximity to special events (e.g., holidays, promotions), where higher values suggest closer proximity.
- **month:** Month in which the session occurred.

- **operating_system**: Operating system used by the visitor.
- **browser**: Browser used during the session.
- **region**: Geographic region of the user.
- **traffic_type**: Source of the traffic (e.g., direct, referral, advertising channels).
- **visitor_type**: Type of visitor (e.g., new or returning user).
- **is_weekend**: Indicates whether the session occurred on a weekend (1) or not (0).
- **engagement_score**: A score reflecting the overall level of user interaction and involvement during the session. Higher values typically indicate deeper browsing activity and stronger interaction with the website.
- **abandonment_score**: A measure associated with the likelihood that a user disengages from the session before completing meaningful actions. Higher values indicate stronger signals of early exit or reduced engagement.
- **high_intent** (*target variable*): Indicates whether the session is classified as a high-intent session (1) or not (0). High-intent sessions represent users with a strong likelihood of conversion or valuable interaction.

Instructions:

You are given two datasets: 'shopper_train.csv' and 'shopper_test.csv'. You will initially explore and analyze the training dataset using statistical methods and visualization tools, and perform preprocess. Afterward, you should train a classification model on the preprocessed dataset and make a prediction for the attached testing dataset.

You should implement the following sections and add more necessary actions, analytics, and visualizations to enrich your work. All the actions you made should be displayed in the Python notebook and the report should include all the explanations with examples and main results.

Perform data analysis:

- Show the data information, e.g., types of attributes, the attributes values etc.
- Show data statistics, e.g., distribution, skewness, median, and more.
- Show and explain attributes correlations.
- Show and explain at least 3 visualizations that present non-trivial insights derived from the data, e.g., involve relationships, patterns, anomalies, etc.

Perform preprocessing. For each step, explain and justify your choice.

- Data cleaning - check for each one of the problems and take care of them properly, e.g., missing values, inconsistency, outliers etc. Explain how you handled each problem for each feature separately and justify your choice.
- If necessary, add and/or delete attributes and create new ones.
- Data transformation - apply the appropriate methods to the required attributes, e.g., normalization, discretization etc.
- Implement the preprocess on the attached test dataset and describe what adjustments you made.

Perform classification:

- Choose and explain the most appropriate evaluation metrics for this problem. Explain why your chosen evaluation metric is appropriate, considering the dataset properties and the business objective.
- Use cross-validation (e.g., k-fold) to evaluate model performance.
- Train at least three different classifiers, present evaluation metrics to compare them, and describe the results. Include at least one new classifier that wasn't covered in class and explain it briefly.
- Consider the class distribution when choosing classifiers and generating predictions.
- Perform hyperparameter tuning for each of the models you train. Describe how you chose the hyperparameters to calibrate and their range.
- Make sure your models are not overfitting or underfitting.
- Compare the models and explain why one outperforms the other based on their characteristics and the dataset.
- Choose the best model in terms of statistical significance and apply it to the attached test dataset to generate a predicted classification. Save the results in a CSV file. Note, the models will be ranked relative to the class based on accuracy.
- Incorporate visualizations whenever possible.
- **For every preprocessing step and modeling choice, justify your decision based on observations from the dataset.**

Good Luck!